COLLEGE OF ENGINEERING & TECHNOLOGY,
SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
KATTANKULATHUR-603203

Formative Assessment- II

Subject Code: 21IPC501J[2] Subject Title: Research Methodology
Register Number: Name of the student :
Year / SEM / Branch / Class:
Exam Date: 03.12.2024 Time : 10.00 am to 12.00 noon

PART-A (5*10 = 50 Marks)

Answer ANY Five

1. a) **Research design in exploratory studies must be flexible but in descriptive studies it must minimize bias and maximize reliability- Discuss**

	Type of study	
Research Design	*Exploratory of Formulative*	*Descriptive/Diagnostic*
Overall design	Flexible design (design must provide opportunity for considering different aspects of the problem)	Rigid design (design must make enough provision for protection against bias and must maximise reliability)
(i) Sampling design	Non-probability sampling design (purposive or judgement sampling)	Probability sampling design (random sampling)
(ii) Statistical design	No pre-planned design for analysis	Pre-planned design for analysis
(iii) Observational design	Unstructured instruments for collection of data	Structured or well thought out instruments for collection of data
(iv) Operational design	No fixed decisions about the operational procedures	Advanced decisions about operational procedures.

b) **Compare and contrast evolutionary algorithms (EAs) and swarm intelligence-based algorithms in terms of their fundamental principles and mechanisms. Provide examples of each type of algorithm.**

Core Mechanisms

Swarm Intelligence

SI is inspired by the collective behavior of decentralized systems, such as bird flocks or fish schools. The key characteristics include:

- **Decentralization**: There is no central control; each agent operates based on local information.
- **Cooperation**: Agents share information and collaborate to find optimal solutions.
- **Adaptation**: The system adapts to changes in the environment through the interactions of its agents.
- Swarm based algorithms are more accurate and robust than evolutionary algorithms in general.
- Swarm-based optimization algorithms use position-updating for global search.

Evolutionary Algorithms

In contrast, EAs are inspired by the process of natural selection. They typically involve:

- **Population-Based Search**: A population of candidate solutions evolves over generations.
- **Genetic Operators**: Selection, crossover, and mutation are used to create new solutions.
- **Fitness Evaluation**: Each solution is evaluated based on a fitness function, guiding the evolution process.
- Evolutionary algorithms are faster than swarm-based algorithms in terms of CPU running times.
- Evolutionary optimization algorithms are population based meta-heuristics and build on generation-updating and thus natural selection.

Key Differences

1. **Search Strategy**:
   - SI employs a collective search strategy where agents explore the solution space simultaneously,
     often leading to faster convergence.
   - EAs utilize a generational approach, where solutions evolve over time, which can be slower but
     allows for more thorough exploration of the search space.
2. **Information Sharing**:
   - In SI, agents share information about their discoveries, which can lead to emergent behavior and rapid adaptation.
   - EAs rely on the fitness of individuals to guide the evolution process, with less emphasis on direct communication between solutions.
3. **Adaptation Mechanism**:
   - SI adapts through the interactions of agents, allowing for dynamic responses to environmental changes.
   - EAs adapt through genetic operators that modify the population based on fitness evaluations, which can be less responsive to sudden changes.

- **Complex Optimization Problems**: SI may be more effective in scenarios requiring rapid adaptation and exploration, such as dynamic environments.
- **Structured Search Spaces**: EAs might excel in well-defined problems where the fitness landscape is known, allowing for effective exploitation of good solutions.
  Ex:
  1. Mult objective problem where minimization or maximization of an objective function is needed
  2. To find an optimal solution in quality control
     Examples: In segmentation of an image into different regions, In MPPT algorithm for maximum extraction of solar power

2. **A group of 150 college students were asked to indicate their most liked film star from six different well known film actors namely A,B,C,D,E and F in order to ascertain their popularity. The observed data were as follows:**

Actors	A	B	C	D	E	F
Frequencies	24	20	32	25	28	21

Test at 5% whether all actors are equally popular.

2)

Solution:

The probability of obtaining any one of the six number is 1/6. Expected frequency of any one number coming upward is $\boxed{\frac{150}{6} = 25}$ = Expected frequency.

Actors.	Observed Frequency (O_i)	Expected Frequency E_i	$(O_i - E_i)$	$(O_i - E_i)^2$	$(O_i-E_i)^2/E_i$
A	24	25	-1	1	1/25
B	20	25	-5	25	25/25
C	32	25	7	49	49/25
D	25	25	0	0	0/25
E	28	25	3	9	9/25
F	21.	25	-4	16	16/25
	Total = 150				

$$\Sigma \left[(O_i - E_i)^2 / E_i \right] = 4.$$

Hence, the calculated value of $\chi^2 = 4$.

Degrees of freedom in the given problem is

$$(n-1) = (6-1) = 5$$

Inference: the table value of χ^2 for 5 degree of freedom at 5% level of significance is 11.071. the result, thus supports the hypothesis and it can be concluded that the die is unbiased.

3. **The surface finish of products produced in a machine shop is suspected to be affected by factors operator and machine. The data of this experiment with two replications in different treatment combinations are summarized in the given table. Analyze and check the significance of components of the related model when significance level is 0.05**

		Machine (B)	
		1	2
Operator (A)	1	65 70	20 40
	2	30 35	50 40

3. Solution

Let the Factor operator be A and the another Factor machine be B. all the Factors assumed as Fixed factors. Number of levels $n=2$. Number of replication in each treatment combination $k=2$, significance level $\alpha = 0.05$

$$Y_{ijk} = \mu + A_i + B_j + AB_{ij} + e_{ijk}$$

		Machines (B) 1		2	
Operator (A)	1	65 70	1 $\boxed{135}$	20 40	b $\boxed{60}$
	2	30 35	a $\boxed{65}$	50 40	ab $\boxed{90}$

$$\text{Sum of squares } A = \frac{(\text{contrast } A)^2}{4k} = \frac{(-1+a-b+ab)^2}{4\times 2}$$

$$= \frac{(-135+65-60+90)^2}{8}$$

$$= 200.$$

$$\text{Sum of squares } B = \frac{(\text{Contrast } B)^2}{4k} = \frac{(-1+a+b+ab)^2}{4\times 2}$$

$$= \frac{(-135-65+60+90)^2}{8}$$

$$= 312\cdot 5$$

$$\text{Sum of squares of } AB = \frac{(\text{Contrast } AB)^2}{4k} = \frac{(1-a-b+ab)^2}{4\times 2}$$

$$= \frac{(135-65-60+90)^2}{8}$$

$$= 1250.$$

4. a) **Explain the assumptions underlying the Chi-Square test for independence. How do violations of these assumptions affect the validity of the test? (4)**

- The data at hand must have been randomly selected (to minimize potential biases) and the variables in question must be nominal or ordinal (there are other methods to test the statistical independence of interval/ratio variables
- The observations between groups should be independent, which basically means the groups are made up of different people
- The data in the cells should be frequencies, or counts of cases rather than percentages or some other transformation of the data.
- The levels (or categories) of the variables are mutually exclusive. ...
- Each subject may contribute data to one and only one cell in the χ^2.
- While Chi-square has no rule about limiting the number of cells (by limiting the number of categories for each variable), a very large number of cells (over 20) can make it difficult to meet assumption #6 below, and to interpret the meaning of the results.The value of the cell *expecteds* should be 5 or more in at least 80% of the cells, and no cell should have an expected of less than one

Violations change the conclusion of the research and interpretation of the results.

- The Chi-Square Test can give misleading results if some cells in the contingency table have very low frequencies or are empty
- The Chi-Square Test is not robust against missing data
- small samples can lead to problems with the chi-square approximation. Also, low expected frequencies in the cells of the contingency table (less than 5) can make the test less reliable.
- it doesn't provide information about the strength or direction of this association.
- Improper categorization can lead to loss of information and potential bias

b) **An oil company was interested in testing four different blends of gasoline for fuel efficiency by controlling the variability of four different drivers and four models of cars. The fuel efficiency was measured as kilometer per litre after driving the cars over a standard clause. Analyze the data and write the inference based on the results.**
[given F(0.05)] =4.76 for (3,6) df] (6)

A(13)	D(9)	C(10)	B(12)
B(12)	C(10)	A(13)	D(8)
C(10)	B(12)	D(9)	A(13)
D(9)	A(14)	B(12)	C(10.5)

Solution

No.of Rows, R = 4
No.of Columns , C = 4

No. of observation, N (Rows $\times$ Columns) = $R \times C$

A(13)	D(9)	C(10)	B(12)
B(12)	C(10)	A(13)	D(8)
C(10)	B(12)	D(9)	A(13)
D(9)	A(14)	B(12)	C(10.5)

= 4×4 = 16

A(13)	D(9)	C(10)	B(12)	Σy_{ij}
B(12)	C(10)	A(13)	D(8)	
C(10)	B(12)	D(9)	A(13)	
D(9)	A(14)	B(12)	C(10.5)	
				176.5

Grand Total, $T = \Sigma y_{ij}$ = 176.5
Correction Factor , C. F = $(\Sigma y_{ij})^2/N$ = $175.5^2/16$
= 1947.02

A(169)	D(81)	C(100)	B(144)	Σy_{ij}^2
B(144)	C(100)	A(169)	D(64)	
C(100)	B(144)	D(81)	A(169)	
D(81)	A(196)	B(144)	C(110.25)	
494	521	494	487.25	1996.25

Total Sum of Square, TSS = Σy_{ij}^2 - C. F
= 1996.25 - 1947.02
= 49.23

					R_i	R_i^2	
	A(13)	D(9)	C(10)	B(12)	44	1936	
	B(12)	C(10)	A(13)	D(8)	43	1849	
	C(10)	B(12)	D(9)	A(13)	44	1936	
	D(9)	A(14)	B(12)	C(10.5)	45.5	2070.25	
C_j	44	45	44	43.5		7791.25	(ΣR_i^2)
C_j^2	1936	2025	1936	1892.25	7789.25		
					(ΣC_j^2)		

Sum of Square between Rows, SSR = $(\Sigma R_i^2)/C$ - C. F
= 7791.25/4 - 1947.02
= 1947.81 - 1947.02

= 49.23

					R_i	R_i^2	
	A(13)	D(9)	C(10)	B(12)	44	1936	
	B(12)	C(10)	A(13)	D(8)	43	1849	
	C(10)	B(12)	D(9)	A(13)	44	1936	
	D(9)	A(14)	B(12)	C(10.5)	45.5	2070.25	
C_j	44	45	44	43.5		7791.25	$(\sum R_i^2)$
C_j^2	1936	2025	1936	1892.25	7789.25		
					$(\sum C_j^2)$		

Sum of Square between Rows, SSR $= (\sum R_i^2)/C - C.F$
$= 7791.25/4 - 1947.02$
$= 1947.81 - 1947.02$
$= 0.79$

Sum of Square btn Columns, SSC $= (\sum C_i^2)/R - C.F$
$= 7789.25/4 - 1947.02$
$= 1947.31 - 1947.02$
$= 0.29$

Analysis:

Source of Variation	Degree of Freedom	Sum of Square	Mean Sum of Square	Observed Value	Table Value
	DF	SS	MSS = SS/DF	Fo = MSS/MSE	Fe =F(,E)
Between Rows	R-1 = 3	0.79	MSR=SSR/(R-1) = 0.2633	0.2633/0.1017 = 2.59	F(3,6)= 4.76
Between Columns	C-1 = 3	0.29	MSC=SSC/(R-1) = 0.0967	0.0967/0.1017 = 0.95	F(3,6)= 4.76
Between Treatmemt	T-1 = 3	47.54	MST=SST/(T-1) = 15.8467	15.8467/0.1017 = 155.82	F(3,6)= 4.76
Error	N-DFR-DFC-DFT= 6	0.61	MSE=SSE/Error = 0.1017		
Total	N-1 = RC-1 = 15				

Inference:

Rows	Columns	Treatment
F0 = 2.59 Fe = 4.76 Since Fo < Fe, we infer that there is no significant difference between Rows.	F0 = 0.95 Fe = 4.76 Since Fo < Fe, we infer that there is no significant difference between Columns.	F0 = 155.82 Fe = 4.76 Since Fo > Fe, we infer that there is significant difference between Treatments.

5. a) **A factory produces items with a defect rate of 2%. If a sample of 50 items is selected, calculate the probability of finding exactly 3 defective items using the Binomial distribution.**

They find out that three in the sample are defective. 2% of the supplier's output is defective
2% of 50 = 1
$P(r = 2) = {}^{n}C_{x}\, p^{x}\, (q)^{n-x}$

5. a)

Given data

$n = 50 \longrightarrow$ no of items

$p = 2\% \Rightarrow p = 0.02$

$q = 1 - p \Rightarrow 0.98$

To find

Binomial distribution for exactly 3 defective items

soln

Formula used

$$P(x = 3) = {}^{n}C_{x}\, p^{x}\, q^{n-x} \qquad \because {}^{n}C_{x} = \frac{n!}{x!(n-x)!}$$

$$= {}^{50}C_{3}\,(0.02)^{3}\,(0.98)^{47}$$

$$= \frac{50!}{3! * 47!}\,(0.02)^{3}\,(0.98)^{47}$$

$$= 19600 * (0.02)^{3} * (0.98)^{47}$$

$$= 19600 * 0.0004 * 0.00656$$

$$\boxed{P(x = 3) = 0.05}$$

b) **The number of customer arrivals at a service desk follows a Poisson distribution with a mean of 5 per hour. What is the probability that exactly 7 customers will arrive in the next hour?**

5. b.

Given data

mean of 5 per hour $\delta = 5$

soln

poisson distribution for exactly 7 customers

formula used $\longrightarrow \dfrac{e^{-\delta} \delta^{x}}{x!}$

$P(x = 7, 5) \Rightarrow$

$$= \frac{e^{-5} 5^{7}}{7!}$$

$$= \frac{0.006 \times 78125}{7 \times 6 \times 5 \times 4 \times 3 \times 2}$$

$$= \frac{0.006 \times 78125}{5040}$$

$$\boxed{P(x = 7, 5) = 0.093}$$

6. a) **A clinical trial involves 100 patients, where the probability of side effects from a medication is 0.01. Using an appropriate distribution, estimate the probability that more than 2 patients will experience side effects. Justify your choice of distribution. [4]**

6. a.

Given data

$n = 100 \quad p = 0.01 \qquad {}^nC_x = \frac{n!}{x!\,(n-x)!}$

$q = 1 - P \quad q = 0.99$

Soln

Binomial distribution for more than 2 patients

formula used

$$P(X \geq 2) = 1 - \sum_{X=0}^{2} {}^nC_x\, p^x q^{n-x}$$

$$= 1 - \sum_{x=0}^{2} nc_x\, p^x q^{n-x}$$

$$= 1 - (nc_0\, p^0 q^{n-0}) + (nc_1\, p^1 q^{n-1}) + (nc_2\, p^2 q^{n-2})$$

$$= 1 - \left[100c_0\,(0.01)^0 (0.98)^{100} + 100c_1\,(0.01)^1 (0.98)^{99} + 100c_2\,(0.01)^2 (0.98)^{98}\right]$$

$$= 1 - \left[0.1326 + 100 \times 0.01 \times 0.135 + 4950 \times 0.0001 \times 0.138087\right]$$

$$= 1 - \left[0.1326 + 0.135 + 0.06831\right]$$

$$= 1 - 0.33591$$

$$\boxed{P(x \geq 2) = 0.6641}$$

b) **A manufacturing process involves multiple controllable factors affecting product quality. Describe how you would use Taguchi's orthogonal array to determine the optimal settings. Provide a specific example.** **[6]**

- Taguchi's orthogonal arrays are highly fractional orthogonal designs. These designs can be used to estimate main effects using only a few experimental runs.
- The L4 array is denoted as L4(2^3).
- L4 means the array requires 4 runs. 2^3 indicates that the design estimates up to three main effects at 2 levels each. The L4 array can be used to estimate three main effects using four runs provided that the two factor and three factor interactions can be ignored.
- Taguchi experiments **often** use a **2-step optimization process**.
- **In step 1** use the signal-to-noise ratio to identify those control factors that reduce variability.

- **In step 2**, identify control factors that move the mean to target and have a small or no effect on the signal-to-noise ratio.
- **Ex:** A microprocessor company is having difficulty with its current yields. Silicon processors are made on a large die, cut into pieces, and each one is tested to match specifications.

- The company has requested that you run experiments to increase processor yield. The factors that affect processor yields are temperature, pressure, doping amount, and deposition rate.

    - The operating conditions for each parameter and level are listed below:

    - **A: Temperature**
        - A1 = 100ºC
        - A2 = 150ºC (current)
        - A3 = 200ºC
    - **B: Pressure**
        - B1 = 2 psi
        - B2 = 5 psi (current)
        - B3 = 8 psi
    - **C: Doping Amount**
        - C1 = 4%
        - C2 = 6% (current)
        - C3 = 8%
    - **D: Deposition Rate**
        - D1 = 0.1 mg/s
        - D2 = 0.2 mg/s (current)
        - D3 = 0.3 mg/s

- The L9 orthogonal array should be used

Experiment Number	Temperature	Pressure	Doping Amount	Deposition Rate	Trial 1	Trial 2	Trial 3	Mean	Standard deviation
1	100	2	4	0.1	87.3	82.3	70.7	80.1	8.5
2	100	5	6	0.2	74.8	70.7	63.2	69.6	5.9
3	100	8	8	0.3	56.5	54.9	45.7	52.4	5.8
4	150	2	6	0.3	79.8	78.2	62.3	73.4	9.7
5	150	5	8	0.1	77.3	76.5	54.9	69.6	12.7
6	150	8	4	0.2	89	87.3	83.2	86.5	3
7	200	2	8	0.2	64.8	62.3	55.7	60.9	4.7
8	200	5	4	0.3	99	93.2	87.3	93.2	5.9
9	200	8	6	0.1	75.7	74	63.2	71	6.8

Experiment Number	A (temp)	B (pres)	C (dop)	D (dep)	SNi
1	1	1	1	1	19.5
2	1	2	2	2	21.5
3	1	3	3	3	19.1
4	2	1	2	3	17.6
5	2	2	3	1	14.8
6	2	3	1	2	29.3
7	3	1	1	2	22.3
8	3	2	2	3	24.0
9	3	3	1	1	20.4

$$SN_{B1} = \frac{19.5 + 17.6 + 22.3}{3} = 19.8 \qquad SN_{B2} = \frac{21.5 + 14.8 + 24.0}{3} = 20.1$$

$$SN_{B3} = \frac{19.1 + 29.3 + 20.4}{3} = 22.9$$

Level	A (temp)	B (pres)	C (dop)	D (dep)
1	20	**19.8**	24.3	18.2
2	20.6	**20.1**	19.8	24.4
3	22.2	**22.9**	18.7	20.2
Δ	2.2	3.1	5.5	6.1
Rank	4	3	2	1

The effect of this factor is then calculated by determining the range:
$\Delta = Max - Min = 22.9 - 19.8 = 3.1$

Deposition rate has the largest effect on the processor yield
and the temperature has the smallest effect on the processor yield.

7. a) **Differentiate between an independent samples T-test and a paired samples T-test. In what scenarios would you use each?**

Paired-samples t tests compare scores on two different variables but for the same group of cases; independent-samples t tests compare scores on the same variable but for two different groups of cases.

Ex:

Comparing students' average quiz scores before they took a study break and after they took a study break. In this case, there exists one group with two conditions: before taking a break and after taking a break. The paired t-test is designed to compare these two groups of scores. Independent t-test compares the means of two independent groups or items. For example, To find any difference in average quiz scores between male and female students. In that case, Independent t-test is used to compare their scores without assuming that there are any differences in variance between genders.

b) **A company claims that the mean lifetime of its batteries is 100 hours. A sample of 25 batteries has a mean lifetime of 98 hours with a standard deviation of 5 hours. Perform a one-sample T-test to evaluate the company's claim at a 5% significance level.**

$H_0 : \mu_{H_0} = 100$ hereX_bar = 98,

$H_a : \mu_{H_0} \neq 100$ mew =100

$$t = \frac{\bar{X} - \mu_{H_0}}{\sigma_s / \sqrt{n}}$$

sd = 98, n=25

Substituting in the above formula

t = -2

Degree of freedom =n-1=24

As per table for df=24 and at a 5% significance level , t _table =2.064
Observed value or t observed < table value or expected value

Rule to make inference :-

If $t_o < t_e$, we infer that training is not effective	If $t_o > t_e$, we infer that training is effective

So hypothesis is rejected